

Lexitarian Population Ethics: A Position-Based Axiomatic Approach*

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Abstract

The paper makes three contributions to the preference aggregation and population ethics literatures. First, we propose the “lexitarian” criterion, a generalization of utilitarianism, according to which the social planner maximizes a vector of lexicographically-ranked utilitarian functions. Second, we suggest modeling the social choice problem using “positions” as a primitive, and axiomatize the lexitarian rule in this context. Third, we argue for an interpretation of the model in terms of the sympathetic paradigm, while prioritizing needs over wants.

1 Introduction

1.1 Motivation

What makes a society “good”? Would we deem it an improvement to impose a modest decrease in the well-being of 99% of society’s members in order to attain immense gains for the remaining 1%? Is a small share of starving children worth broad access to education for the remaining majority? What if education is replaced by exposure to opera? These problems become even more complicated when we relax the constraint on a constant population size (cf. Parfit, 1976, 1982): do we prefer a population of 10 billion people living in

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comfort, or 20 billion who suffer recurring hunger? Yet more difficulties arise when discussing future populations. How do we evaluate the welfare of people whose tastes and circumstances are, at best, conjectures?

It is tempting to avoid these difficulties by insisting that individual utilities are not commensurate and then seeking refuge in the Pareto ranking. This is the path taken by much of economic theory. However, decisions have to be made, even when alternatives are not Pareto ranked, and it would often be useful to have a workable idea of social welfare in settings with non-trivial interpersonal trade-offs.

In this paper, we propose a model of welfare aggregation for fixed or variable populations. We make three contributions. First, we propose a “lexitarian” rule, according to which there are various priorities, captured by several utility functions, reflecting various priorities, and the social planner attempts to maximize their sums in a lexicographic manner. This generalization of utilitarianism captures the intuition that trade-offs between individuals’ welfare are inevitably considered in public policy and social planners’ problems, but that not all sources of suffering or happiness are created equal. Some trade-offs may be deemed outright unpalatable while others remain a necessary compromise. Second, we introduce the notion of a “position” as the basic primitive, and consider social profiles that capture the number of individuals in each position, rather than the experiences of each individual per se. Finally, we interpret our model in terms of the sympathetic paradigm of ethics, while focusing on needs rather than wants. The next three subsections develop each of these points.

1.2 The Aggregation: Lexitarianism

The summation of individual measures of well-being as a measure of aggregate social welfare is intuitively very appealing. It also frequently makes sense to inconvenience a small number of people for the greater good of a society. At the same time, we often have the intuition that there are some sacrifices we will not ask, even on the part of a few, for the advantage of the many. To cope with this tension while retaining the appeal of utilitarian aggregation, we

propose a lexicographic generalization of utilitarianism.

Under the standard utilitarian approach to social welfare, each of the members of a population K is characterized by an outcome drawn from a set P of measures of well-being. Aggregate well-being is given by $\sum_{i \in K} u_i(p_i)$, where u_i is person i 's utility function. Section 1.4 explains that we adopt a sympathetic framework in which the social planner uses a single utility function u to evaluate the outcome for every member of the population, so that the utilitarian aggregation takes the form

$$U(p_1, \dots, p_k) = \sum_{i \in K} u(p_i). \quad (1)$$

Now assume that the social planner has several functions $u_j : P \rightarrow \mathbb{R}$ with $j = 1, \dots, m$ and consider the lexicographic ordering induced by successively calculating (1) for the functions $(u_1, \dots, u_m)$. This enables us to represent preferences that refuse to make trade-offs between certain needs or wants. For example, we can imagine the function u_1 as measuring health and nutrition and u_2 measuring self-fulfillment and personal expression. Such a model would allow one to say that providing medication to sick people may justify lowering the level of nutrition of some others, but that art classes for some individuals should not be traded for the nutrition of others.

1.3 The Framework: Positions

When discussing population ethics, in particular when considering the far future, we find it more intuitive to think of the numbers of people who will live in certain conditions rather than of their individual identities. We accordingly propose a model in which the primitive is a set of “positions” people may occupy. The objects of choice are population profiles, which describe how many individuals are at each position. Formally, a profile is a “counter vector” I , where $I(p) \geq 0$ indicates how many individuals are at position p . The overall number of individuals in a profile (ranging over all positions) is finite, but we impose no a priori bounds—neither on the number of positions which may be occupied, nor on the number of individuals at each position.

It will be helpful to compare the position-based model to the standard person-based approach. Assume, as above, a population $K = \{1, \dots, k\}$ of individuals with outcomes drawn from a set P , and a sympathetic utility function $u : P \rightarrow \mathbb{R}$. The standard person-based approach considers the objects of choice to be functions $f : K \rightarrow P$, deriving the objective $U(p_1, \dots, p_k) = \sum_{i \in K} u(p_i)$ from rankings over such functions f . By contrast, in our position-based model the objects of choice are functions whose domain is P and range is the nonnegative integers, i.e., $I : P \rightarrow \mathbb{Z}_+$, so that $I(p)$ identifies the number of people characterized by outcome p . Given a utility function $u : P \rightarrow \mathbb{R}$, the planner maximizes

$$I \cdot u = \sum_{p \in P} I(p) u(p).$$

With enough freedom in the definition of a position, the position-based and person-based approaches are equivalent. Given a function $f : K \rightarrow P$, we may define a profile I by $I(p) = |f^{-1}(p)|$ so that $I \cdot u = U(p_1, \dots, p_k)$. Conversely, given a profile $I : P \rightarrow \mathbb{Z}_+$, we can construct a population $K = \{1, \dots, k\}$ for $k = \sum_{p \in P} I(p)$ and a function $f : K \rightarrow P$ such that, for each p with $I(p) > 0$, there are $I(p)$ identical individuals i with $f(i) = p$, so that $U(p_1, \dots, p_k) = I \cdot u$. We can extend this reasoning to obtain an equivalence between hierarchies of utility functions ranking alternatives $f : K \rightarrow P$ and hierarchies of utility functions ranking alternatives $I : P \rightarrow \mathbb{Z}_+$. Lexicographic person-based and position-based models are thus formally equivalent.

The advantage of the position-based model is that it offers a useful new perspective on social welfare. First, as will be clear below, the axiomatic derivation of the model suggests a new way of conceptualizing utilitarianism and its variants. Specifically, the lexitarian model will be simply axiomatized, with the lexicographic order over utility functions naturally derived from the social planner's preferences. Second, a specific position can also be interpreted as capturing only a part of a person's experience. Thus, in our model each person might be at several positions, reflecting different needs, states of the world, or time periods. This allows us to encompass cases in which some competing needs and wants are commensurate while others are not. Lastly,

the positions model has a built-in anonymity assumption: only the numbers of individuals (in each position) matter, not their identities.

1.4 The Sympathetic, Impersonal Approach

Choice of population: A philosophical schism divides the study of population ethics into two main fields of thought: person-affecting views and impersonal approaches (see, e.g., Carlson (1998), Arrhenius (2000), Roberts (2011), Boonin (2014), and Frick (2020); the issue has also been discussed in social choice theory by Blackorby and Donaldson (1984), among others). Person-affecting theories (see, e.g., Navreson, 1973, for a foundational work) maintain that an outcome can only be better or worse if it is better or worse for particular individuals across the compared alternatives, giving rise to the “Non-Identity Problem” and other difficulties in evaluating the moral significance of creating new lives (Parfit, 1984).¹ Lexitarianism, with its flexibility in considering (varying) outcome-specific populations, is more aligned with the impersonal view.

Parfit (1984) raises another common critique to which aggregative welfare approaches—particularly those of the impersonal paradigm—are exposed: the “Repugnant Conclusion” observes that a sufficiently-large, marginally-subsistent society would always be mechanically preferred to a small, flourishing one.² Several directions have emerged in the population ethics literature to address this critique, including variants of average utilitarianism³ (Harsanyi,

¹The person-affecting view is famously able to accommodate the asymmetry of contingent-existence: “we are in favour of making people happy,” wrote Navreson (1973), “but neutral about making happy people”. The Non-Identity Problem (Parfit, 1984), conversely points out that it can be appropriate to consider the beneficiaries or victims whose existence is conditional on the action taken (e.g., if evaluating aggressive resource-depletion programmes).

²Asheim and Cato (2026) show that the Repugnant Conclusion, also known as the “Mere Addition Paradox” and the related “Very Sadistic Conclusion” are quite general, and do not depend on comparability assumptions such as anonymity or Archimedeanity. The paradoxical nature of the problem is contested: some philosophers have disputed the “repugnancy” of the conclusion (Huemer, 2008; Tännsjö, 2002; Gustaffson, 2022), while others have considered this as revelatory of a fundamental flaw to aggregation of welfare in ethics and utilitarianism (Temkin, 2012).

³Which, nonetheless, are exposed to critiques analogous to the repugnant conclusion; see

1955; Sikora, 1975); rank-dependent approaches (Zuber and Asheim, 2012); prioritarianism (Adler, 2012; Parfit, 1997); and various threshold-based modifications, including leximin principles (Hammond, 1976), utilitarianism dampened by population (Hurka, 1983), maximin welfare orderings (Bossert, 1990), and critical thresholds below which individuals’ well-being does not contribute to aggregate welfare (Blackorby, Bossert, and Donaldson, 1995; 1997). Recent contributions that deal with the repugnant conclusion and related problems in a utilitarian framework include Pivato (2020) and Nebel (2024).

Our model nests the generally “prioritarian” impulse of focusing on the most-suffering and the threshold-based intuition, common to these approaches, and extends it by including a rank-ordering of priorities on which the social planner lexicographically judges a society.

Choice of utility function: In preference aggregation, two natural choices are possible. One may defer in the choice of utility function to the individuals, such that each is accorded a measure of their own welfare, or one may insist on a standard universal choice of utility function. Adopting the language of Bloom (2016), we view the distinction as motivated by empathetic and sympathetic considerations: a social planner may empathetically attempt to “experience the world as [he] thinks someone else does,” or sympathetically care and wish the best for others, without nonetheless fully adopting their perspective.⁴ The empathetic view appears throughout the economics literature in the form of phrases such as “de gustibus non est disputandum” or “consumer sovereignty”, and in models of utilitarianism such as Harsanyi (1955) and Dhillon and Mertens (1999).

The lexitarian model is written within the sympathetic paradigm. Furthermore, we find the model’s assumptions more palatable if positions identify

also Parfit (1984) and Arrhenius (2000).

⁴The distinction goes back, at least, to Harsanyi (1955) and – though different in language and normative in flavour – to Mill (1863). The latter argued that even with different cognitive capacities and understandings of pleasure, it is “better to be Socrates dissatisfied than a fool satisfied”, or, in other words, that idiosyncratic constraints on perception or understanding of welfare do not obviate the need for (or indeed the validity of) an objective evaluation of well-being (Mill, 1863, Ch. 2).

needs and the degree to which they are satisfied rather than preferences (or wants). First, individual preferences are difficult to observe and, given incentives to misrepresent them, difficult to elicit. As a result, attempts to aggregate individuals' "true" subjective utilities, in an empathetic fashion, are likely to be of limited success. Second, even if these utility functions were directly and unambiguously observable, maximizing their sums would, as we argue, give rise to unacceptably counterintuitive implications (see section 4.1). Third, we expect to have a better idea of the needs of future generations than of their wants. Hence, when considering these generations, and also when trading-off their well-being against that of the present generation, it seems sounder to rely on needs rather than on wants. Finally, if one follows the sympathetic approach in allowing a social planner to gauge the utility of individuals based on criteria she deems salient, without soliciting the individuals' subjective assessments of their own positions, then one may wish to limit that social planner's power. Restricting their discourse to perceived needs may be an important step in this direction.

The lexitarian aggregation of preferences is designed to identify determinants of well-being that can be compared and traded-off, and to distinguish them from those pairs that cannot. As such, our model relates to several lines of thought in philosophy and in psychology. One of the most prominent contributions is Rawls's (1971) theory of justice, which suggests that a social profile be evaluated by the welfare of the individuals who are worst-off. The resulting maxmin rule suggests that the central planner should focus on unsatisfied basic needs before considering less-basic ones, let alone desires that are considered to be in the domain of wants, or luxury. Indeed, this approach can be modeled in our framework, using an infinite hierarchy of utility functions, $(u_\alpha)_\alpha$, each indicating whether the individual in question obtains a level of welfare α . In such a model one recovers the leximin criterion (Osborne, 2025, Section 1.8), which refines the maxmin one. However, the maxmin (or leximin) criterion is often considered extreme. For example, if the poorest individual has to survive on 400 grams of rice a day, no amount of food given to millions of others would be considered an aggregate improvement, let alone

justify dropping this quantity to 390 grams a day. Many would view this as somewhat exaggerated. Our model allows for a utilitarian trade-off between quantities of food, without necessarily trading off space trips against food.

Our notion of a hierarchy of needs can generalize sufficientarian (see Frankfurt, 1987, Alcantud, Mariotti, and Veneziani, 2022, and Bossert, Cato, and Kamaga, 2023). Sufficientarianism suggests that ethical evaluations should consider the number of individuals who “have enough”, without taking into consideration how much more they can have, once they have reached the “sufficiency” threshold. We can capture this in our model by letting utility be an indicator function of “having enough”. We can then allow for refinements, both below and above the threshold. For example, a function u_1 can be an indicator, stating whether an individual has or has not reached a “sufficient” level of a resource, and u_2 may capture other needs or wants the individual might have. Thus, a lexitarian model can respect people’s preferences for, say, intellectual stimulation, while insisting that such concerns do not enter the picture until everyone has reached a minimum level of nourishment. In this sense, Maslow’s (1943) hierarchy of needs can be a source of inspiration for the social planner’s lexicographic preferences. In fact, one may view our model as a Maslowian version of Bentham’s utilitarian theory.

1.5 Planners and Theorists

The interpretation of the model may vary with the context and the planner’s identity. The social planner may be a legislative body who makes decisions on general principles of social and economic policies, or a government agency who implements such general policies. In some cases, one may imagine a planner being a parent making decisions that will affect her children. Alternatively, the planner may be a rhetorical device that a policy advocate uses in arguing for the merits of her policy. The main application we have in mind, however, involves a planner who is called upon to make decisions about economic resources. As such, we imagine the planner’s choices as an attempt to capture a social consensus, embodying a society’s ethos as to how policies are to be evaluated.

The discussion in this paper, as, indeed, in most of the literature, is normative in essence. In this context it is useful to bear in mind that discussions of this type often have two types of audiences: fellow economists, who may use models to make policy recommendations, and individuals in society who presumably vote on social policy issues. Offering an axiomatic derivation of a formal model—such as lexitarianism—is mostly addressed at professional economists. Discussing the preferred interpretation of the model—such as the priority of basic needs over wants—should have a more general appeal. Naturally, these distinctions are fuzzy. Moreover, some of the claims can be interpreted in either way. Yet, we find the distinction useful to bear in mind, and we emphasize that any normative claim is to be read as an attempt to capture a wide consensus—whether of professionals or of laypeople.

1.6 Preview

Section 2 describes the model and presents the central result, as well as the derivation of the special case of classical utilitarianism in our framework. Section 3 extends the model to encompass time and uncertainty, decomposing the notion of “position” into circumstances, time-periods, and states.

There are several questions that may be raised about our model of lexitarianism and its suggested interpretations. Specifically, a sympathetic reader may ask:

(i) Suppose we accept the axioms and the model. Why should a central planner ignore individual utilities $(u_i)_i$?⁵

(ii) Next, even if we agree on the sympathetic approach, why should the planner focus on needs rather than wants, and, perhaps more fundamentally, what is the difference between these? Is there an observable, scientific way to tell them apart?

(iii) Finally, at a more technical level, what is the right way to capture

⁵This question is particularly relevant in the case of classical utilitarianism. When general lexitarianism is considered, it is not entirely obvious whether individuals have lexicographic preferences and how these should be aggregated, especially when different individuals might well have different numbers of lexicographically-ordered categories.

needs in a position-based model?

Section 4.2 deals with these questions. It argues that the interpretation and modeling choices presented here offer a coherent approach to modeling population ethics, while recognizing that different readers may accept some of these choices but reject others. We conclude with a general discussion in Section 5, while proofs of the representation theorems are given in the appendix.

2 Model and Results

Let there be a set of *positions* P . The set of *profiles* is

$$\mathcal{I} = \left\{ I : P \rightarrow \mathbb{Z}_+ \mid \sum_{p \in P} I(p) < \infty \right\}.$$

For $I \in \mathcal{I}$, the *support* of I is $\text{supp}(I) \equiv \{p \in P \mid I(p) > 0\}$. Thus, the set of positions P may be infinite, but each profile I has a finite support. Algebraic operations are performed on $\mathcal{I}$ pointwise so that $nI + mJ$ is well-defined for $n, m \geq 0$ and $I, J \in \mathcal{I}$. $0 \in \mathcal{I}$ has its natural meaning. Similarly, for $I, J \in \mathcal{I}$ the inequality $I \geq J$ is read pointwise. For $p \in P$, $\mathbf{1}_p \in \mathcal{I}$ denotes the indicator function for p . For a subset of positions $A \subset P$, we denote by $I_A \in \mathcal{I}$ the corresponding profile that vanishes outside A , i.e. $I_A = \sum_{p \in A} I(p) \mathbf{1}_p$.

For a function $u : P \rightarrow \mathbb{R}$, denote

$$I \cdot u = \sum_{p \in P} I(p) u(p)$$

which is well-defined as I has a finite support. For $u : P \rightarrow \mathbb{R}$ and $A \subset P$, $u_A : A \rightarrow \mathbb{R}$ is the restriction of u to A .

Let there be given a binary relation $\succsim \subset \mathcal{I} \times \mathcal{I}$. The symmetric and asymmetric parts of $\succsim$ will be denoted, as usual, by $\sim$ and $\succ$, respectively.

We impose four axioms. First,

A1 Weak Order: $\succsim$ is complete and transitive.

The completeness axiom is notoriously problematic, especially in the context of social choice. Its main justification is, as always, that decisions need to

be made, and that it is better to bring them forth and see them in the model rather than to leave them unspecified. We return to this point in subsection 4.2.⁶

Transitivity is one of the most basic axioms of choice models, and we do not have anything to say about it that is specific to the current model. The main reason to highlight it is that some of the literature on the “Repugnant Conclusion” suggests dropping this axiom (see Temkin, 1987, Persson, 2004, and Rachels, 2004). We tend to believe that giving up on transitivity is a high price to pay, both theoretically and practically, and that it is worthwhile to explore the possibilities that respect this axiom.

Our next axiom is the main driving force behind the additive representation:

A2 Union: For all $I, J, K \in \mathcal{I}$, $I \succsim J$ iff $I + K \succsim J + K$.

Consider first an interpretation of the model where a position summarizes all the information about an individual, so that each individual is counted in a profile at one position only. In this case, suppose that the social planner can choose between one policy, leading to a population profile I , and another, which will result in a population profile J . There is also another location, where the population profile will be K , irrespective of the current policy choice. The Union axiom states that the preference between I and J should not change if each is combined with the population whose profile is K . Clearly, it is very similar to axioms of separability across populations, and to Savage’s P2 (Savage, 1954). It also bears resemblance to von Neumann and Morgenstern’s (1944, 1947) Independence axiom, basically stating that a subpopulation that is common to two profiles can be ignored.

The Union axiom is, however, more questionable when a position is interpreted as describing only some aspects of an individual’s well-being. For example, if we think food and self-fulfillment as incommensurate needs, and if each individual appears in a food-position as well as in a self-fulfillment-position, the axiom states that we ignore inequality in food distribution when

⁶There are papers which opt for quasi-orderings instead, including, for instance Parfit (1982), which introduces indeterminacy via ranges of utility specified for added individuals.

we consider inequality in self-fulfillment. We return to discuss these modeling choices in subsection 4.2.

The third axiom is a monotonicity axiom, indicating that the central planner prefers that any population survives rather than dies:

A3 Monotonicity: For all $I \in \mathcal{I}$, $I \succsim 0$.

This axiom is far from trivial. One of the more influential attempts to avoid the repugnant conclusion is *critical level utilitarianism* (Blackorby, Bossert, and Donaldson, 1995; 1997), which posits the existence of a critical threshold below which individuals' well-being does not contribute to aggregate welfare.⁷ In our model, this would be akin to dropping A3 and allowing the utility function $u(p)$ to assume negative values. Indeed, one can prove such a version of our main result. In such a model, the critical level would be endogenously derived by the value 0 on the utility scale.⁸ However, we take here a more conservative approach according to which the central planner is not supposed to decide that certain lives are not worth living.

For our final axiom, we need the notion of commensurability:

Definition 1 For $p, q \in P$, p and q are commensurate if there exists $k \geq 1$ such that $k\mathbf{1}_p \succsim \mathbf{1}_q$ and $l \geq 1$ such that $l\mathbf{1}_q \succsim \mathbf{1}_p$.

Thus, two positions p and q commensurate if a profile with sufficiently many people at position p is at least as desirable as a profile with a single person at q , and vice versa. Intuitively, the order $\succsim$ allows trade-offs over commensurate positions. Note that the intuition for this definition relies on Monotonicity: it is implicitly assumed that being at each position is a good thing, and thus, unless p and q are incommensurate, a large population at one position should eventually be a better society than a single person at the other. Conversely, should one believe that a position p describes “life not worth living”, a larger

⁷See Broome (2004) for a discussion of the merits and drawbacks of aggregative welfare approaches in general, and non-unique critical thresholds in particular.

⁸If we deal with the special case of utilitarianism, namely, when the lexicographic ordering is trivially defined by a single function, one only needs to drop A3 and obtain the general result. For the counterpart of the lexicographic result, the definition of “commensurate”, which follows, would have to be modified.

population at position p would only make the resulting society worse in the eyes of the central planner.

If positions p and q are not commensurate, given completeness, it must be the case that one is infinitely more desirable than another. For example, being healthy may be considered incommensurably more important than having a sense of self-fulfillment. Importantly, the notion is endogenous: it is defined by the social planner's preference relation $\succsim$. Our model does not dictate which positions are incommensurably more important than others, nor that there be such pairs to begin with. With commensurability thus endogenously-defined, we proceed to state the last axiom.

A4 Restricted Archimedeanity: For all $I, J, K, L \in \mathcal{I}$, if $p, q \in \text{supp}(I) \cup \text{supp}(J) \cup \text{supp}(K) \cup \text{supp}(L)$ are commensurate and $K \succ L$, then there exists $n \geq 0$ such that $I + nK \succ J + nL$.

If we ignore the commensurability condition, the axiom is rather standard: it says that, if profile K is strictly preferred to profile L , then taking the union of any profiles I and J with sufficiently many copies of K and L , respectively, would result in a preference for the I -with- K 's population over the J -with- L 's one. The axiom states that the difference between J and I cannot be infinitely weightier than that between K and L . The fact that it is restricted to some quadruples (I, J, K, L) is at the heart of lexicitarianism: the motivation for our model is precisely the fact that some differences are *not* commensurable. However, the axiom states that incommensurability isn't arbitrary, and it is basically defined by positions: if all the positions involved in (I, J, K, L) are pairwise commensurate—according to the planner's own preferences—then the axiom holds.

Next, we define the notion of numerical representation we seek:

Definition 2 A lexicitarian representation of $\succsim$ is a pair $(\succsim^b, u)$ where $\succsim^b$ is a weak order over P and $u : P \rightarrow \mathbb{R}_+$ such that, for all $I, J \in \mathcal{I}$, $I \succ J$ iff there exists a $\succsim^b$ -equivalence class $A \subset P$ such that

$$I_B \cdot u = J_B \cdot u$$

for every $\succsim^b$ equivalence class for which $B \succ^b A$ and

$$I_A \cdot u > J_A \cdot u,$$

where $\sim^b$ and $\succ^b$ are the symmetric and asymmetric parts of $\succsim$, respectively.

Note that, for every $I, J \in \mathcal{I}$, we have $I_B = J_B = 0$ for all but finitely many $\sim^b$ -equivalence classes B . Hence, if $\succsim$ is complete, this definition also implies that $I \sim J$ iff $I_B \cdot u = J_B \cdot u$ holds for every $\sim^b$ -equivalence class B .

A lexicarian representation $(\succsim^b, u)$ is a rather simple object: it starts with a ranking of positions $\succsim^b$, which can be thought of as “at least as basic as”. It also involves a utility function u over positions. The ranking of two profiles I, J by $(\succsim^b, u)$ is defined by seeking the $\succsim^b$ -highest equivalence class that tells them apart, according to their u values. Note that u values over different equivalence classes never get to be compared. Hence we can think of the representation as if there were different utility functions, each defined for a different equivalence class of positions.⁹ If it so happens that $\succsim^b$ has finitely many equivalence classes, we can also use the more standard lexicographic representation, with a utility function for each.

We can finally state the main result, proved in Section 6.2:

Theorem 1 *$\succsim$ satisfies A1, A2, A3 and A4 if and only if it has a lexicarian representation $(\succsim^b, u)$. Moreover, for every $\sim^b$ -equivalence class B , u_B is unique up to multiplication by a number $\lambda_B > 0$.*

Observe that the statement of the theorem doesn’t specify that $\succsim^b$ is unique. It will be clear from the proof that it is unique when restricting attention to the support of u . Positions that are “null” (i.e., positions p such that $\mathbf{1}_p \sim 0$) can be assigned to various $\sim^b$ -equivalence classes.

The statements of the axioms clearly require consideration of populations of varying sizes. However, the implied representation can just as clearly be used to evaluate different allocations for a population of fixed size.

In the case of a finite set equivalence classes, the representation can take a more familiar form:

⁹For notational simplicity, we use but one function u .

Corollary 1 Assume that $\succsim^b$ has finitely many equivalence classes. Then $\succsim$ satisfies A1, A2, A3 and A4 if and only if there exists $u : P \rightarrow \mathbb{R}_+$ and a partition of P , $(A_1, \dots, A_m)$ ($m \geq 1$) such that $u(p) > 0$ for all $p \notin A_m$, and for all $I, J \in \mathcal{I}$, $I \succsim J$ iff

$$(I_{A_1} \cdot u, \dots, I_{A_m} \cdot u) \geq_L (J_{A_1} \cdot u, \dots, J_{A_m} \cdot u),$$

where $\geq_L$ is the familiar lexicographic ordering of elements of $\mathbb{R}^m$.

Note that this representation coincides with the lexicographic representation of $\succsim$ by $(u_1, \dots, u_m)$ where each u_i vanishes outside A_i . Conversely, given a lexicographic representation by finitely many functions $(u_1, \dots, u_m)$ over a population $K = 1, \dots, k$ of individuals, their utility profiles can be embedded in our model by introducing m equivalence classes of positions, each reflecting the range of u_i for a given i , as explained above for the utilitarian case ($m = 1$). Again, we see that with sufficient flexibility in the formulation of a position, lexicographic person-based and position-based models are equivalent.

A simple and familiar special case is utilitarianism, namely the case in which all positions are commensurable. In that case we may assume a stronger version of the Archimedeanity axiom:

A4* Strong Archimedeanity: For all $I, J, K, L \in \mathcal{I}$, if $K \succ L$, then there exists $n \geq 0$ such that $I + nK \succ J + nL$.

For the statement of this axiom, one need not define commensurability of positions. However, it is obvious that (given the other axioms) any two positions p, q will be commensurate. In other words, given A1-A4, we could also impose the condition that any two positions be commensurate and obtain A4*.

Section 6.1 proves the following:

Theorem 2 $\succsim$ satisfies A1, A2, and A4* if and only if there exists a function

$$u : P \rightarrow \mathbb{R}$$

such that, for all $I, J \in \mathcal{I}$

$$\begin{aligned} I &\succsim J \\ &\text{iff} \\ I \cdot u &\geq J \cdot u \end{aligned} \tag{2}$$

Moreover, in this case u is unique up to multiplication by a positive number.

Furthermore, if the above hold, then A3 holds iff $u \geq 0$.

Theorem 1 in Chapter 3 of Krantz, Luce, Suppes, and Tversky (1971, p. 74) encompasses the main part of Theorem 2 (i.e., without A3 and nonnegativity) within a more general and abstract framework.¹⁰ Our setting is more concrete, and, as a result, the axioms and proof are commensurately simpler.¹¹ Indeed, from a mathematical viewpoint, our argument is similar to de Finetti’s (1931, 1937) derivation of maximization of expected value relative to a subjective probability (see the version described in Gilboa, 2009). In that model, P is a set of states of the world, and a real-valued function from the states to the real line describes a bet. Similar axioms are used to derive a separating hyperplane between the bets that are at least as desirable as zero and those that are not, and an additional monotonicity axiom guarantees that (apart from the trivial case of indifference), the gradient of the hyperplane can be interpreted as a probability over the states. The major conceptual difference between the results is that in de Finetti’s model the utility function is assumed known (represented by the values of each function), and the derived weights are probabilities—whereas in the present model the given values are numbers

¹⁰We thank Jacob Nebel for pointing this out to us. The result is also cited as Proposition 1 in Nebel (2024).

¹¹Krantz et al. (1971) deal with an abstract “extensive structure”, involving a binary operation that need not be commutative, nor even associative. (An axiom named “weak associativity” guarantees, however, that in case associativity fails, the order of operations does not affect preferences. No such axiom is needed in case commutativity fails.) Thus, one of their axioms isn’t needed in our setup, and two others can be simplified. More importantly, the additive operation induces a simple geometric structure, and the proof is basically a separating hyperplane argument. The corresponding proof in Krantz et al. is, naturally, more involved, and relies on representations constructed by “standard sequences” in less structured spaces (including Theorems 4 and 5 in their Chapter 2).

of occurrences of positions, and the utility is derived as the weights assigned to these positions. The difference in interpretation also results in a mathematical difference: in the present model the values of the functions compared are only non-negative integers, rather than any real numbers. Consequently, the proof of this result is more involved than it would be in the real-valued case.

In sum, the four axioms are equivalent to the existence of a number $u(p)$ for each position p , such that profiles are evaluated by taking the summation of $u(p)$ over all positions within each equivalence class, and then lexicographically comparing the resulting vectors of sums. The numbers specified by u are unique up to a positive multiplicative constant.¹² There is no assumption about the utility u as a function of p ; the mathematical derivation uses the space of counter vectors (the $I, J, \dots$) and not the utility values or even the positions.

3 Time and Uncertainty

The model as discussed so far implicitly assumes that there is no uncertainty regarding the outcomes of social policies, and that all occur contemporaneously. This is clearly a very restricted setup. People live in different periods of time, and there is considerable uncertainty about the number of people alive at any time, let alone about the position(s) they will occupy. Indeed, some of the most prominent questions of population ethics have to do with the effect of current policies on the uncertainties faced by future generations.

In this section we re-interpret the notion of a “position” to allow us to deal with such questions based on the analysis above. Specifically, we will assume that there is a set of “circumstances”, playing the role of positions in a one-stage model under certainty, and a position will now be a circumstance experienced at a given state of the world and a given period.

¹²While the set of positions may be infinite, every profile considered has only finitely many individuals, so that the summation of their utilities is well-defined. It is possible, for example, that a position is defined by a level of wealth, and that the set of positions is a continuum. Yet, a given population will only have finitely many individuals, and therefore only finitely many positions p such that $I(p) > 0$.

Formally, let C be a set of circumstances, let $T = \{0, 1, 2, \dots\}$ be a set of time periods, and let Ω be a finite set of states of the world. Assume that $P = C \times T \times \Omega$. It will be convenient to denote subsets of $\mathcal{I}$ defined by supports that are restricted to states or time periods. With a slight abuse of notation that is unlikely to cause any confusion, we define

$$\begin{aligned}\mathcal{I}_t &= \{I \in \mathcal{I} \mid I(c, t', \omega) = 0 \quad \forall t' \neq t\} \\ \mathcal{I}_\omega &= \{I \in \mathcal{I} \mid I(c, t, \omega') = 0 \quad \forall \omega' \neq \omega\} \\ \mathcal{I}_{t,t'} &= \{I \in \mathcal{I} \mid I(c, t'', \omega) = 0 \quad \forall t'' \neq t, t'\}\end{aligned}$$

Define a state ω to be *null* if the value of $I(c, t, \omega)$ does not affect preferences. We impose three additional axioms:

A5 State Invariance: For all non-null $\omega, \omega' \in \Omega$, all $I, J \in \mathcal{I}_\omega$, and $I', J' \in \mathcal{I}_{\omega'}$, if for all $(c, t) \in C \times T$, $I(c, t, \omega) = I'(c, t, \omega')$ and $J(c, t, \omega) = J'(c, t, \omega')$, then $I \succsim J$ iff $I' \succsim J'$.

A6 Time Invariance: For all $t, t' \in T$, all $I, J \in \mathcal{I}_t$, and $I', J' \in \mathcal{I}_{t'}$, if for all $(c, \omega) \in C \times \Omega$, $I(c, t, \omega) = I'(c, t', \omega)$ and $J(c, t, \omega) = J'(c, t', \omega)$, then $I \succsim J$ iff $I' \succsim J'$.

Axioms A5 and A6 are the natural conditions required to have a utility function that is state- and time-independent. They can be stated in other ways as well: first, instead of limiting attention to profiles whose support only involves one state (or one period t), they can be stated for any pairs of profiles I, J, I', J' such that I' and J' are obtained from I and J , respectively, by permuting the corresponding states (or periods). This seemingly stronger requirement will obviously be equivalent to the current formulation in light of the additive structure guaranteed by the results of Section 2.

Clearly, Axioms A5 and A6 consider profiles that do not have the same number of individuals in all states and/or in all time periods. This is in line with our general setup, assuming that profiles with different numbers of individuals can be compared. Indeed, this flexibility of the model seems necessary to discuss some of the problems of population ethics: when considering the

long-term risks of climate change or of nuclear wars, the number of individuals alive will typically differ across time periods and across states of the world.

While A5 and A6 treat uncertainty and time symmetrically, the next axiom is specific to the time dimension:

A7 Dynamic Consistency: For all $t \in T$, all $I, J \in \mathcal{I}_{t,t+1}$, and $I', J' \in \mathcal{I}_{t+1,t+2}$, if for all $(c, \omega) \in C \times \Omega$ and $t' \in T$, $I(c, t' + 1, \omega) = I'(c, t', \omega)$ and $J(c, t' + 1, \omega) = J'(c, t', \omega)$, then $I \succsim J$ iff $I' \succsim J'$.

We start with the (simpler) case of utilitarianism, that is, when all positions are commensurate.

Proposition 1 $\succsim$ satisfies A1, A2, A4* as well as A5, A6, and A7 if and only if there exists a utility function defined on circumstances,

$$\hat{u} : C \rightarrow \mathbb{R}$$

a probability vector $pr \in \Delta(\Omega)$ and a factor $\delta > 0$ such that, for all $I, J \in \mathcal{I}$

$$I \succsim J \Leftrightarrow V(I) \geq V(J) \quad (3)$$

where

$$V(I) = \sum_{\omega \in \Omega} pr(\omega) \sum_{t \in T} \delta^t \sum_{c \in C} I(c, t, \omega) \hat{u}(c)$$

Moreover, in this case $\hat{u}$ is unique up to multiplication by a positive number, whereas pr and δ are unique if $\hat{u}$ is not identically zero.¹³

Corollary 2 Finally, under the above conditions, A3 holds iff u is nonnegative.

Note that the proposition does not ensure that $\delta < 1$. Since only vectors I with finite support are considered, there is no mathematical difficulty in allowing $\delta \geq 1$. It is not difficult to state an assumption that would guarantee $\delta < 1$.¹⁴

¹³A constant zero-valued u corresponds to the case of a trivial relation for which $I \sim J$ for all I, J .

¹⁴Such an assumption could compare a vector I in $\mathcal{I}_t$ with the corresponding vector in $\mathcal{I}_{t+1}$ —but the direction of preference between them would have to depend on the preference between I and 0.

For the general case of lexicarian preferences, a few adaptations are needed. First, we have to ask which pairs of positions are reasonably commensurate. Our basic intuition is that commensurability is a matter of the circumstances, not of time or uncertainty. For example, luxury goods may be incommensurate to life-saving medication, but there is no reason that luxury goods ten years from now would be incommensurate to the same goods today, or that hunger in the state of global warming would be incommensurate to hunger in the state of no climate change. We therefore impose the following:

A8 Commensurate: For all $c \in C$, $(t, \omega), (t', \omega') \in T \times \Omega$, (c, t, ω) and (c, t', ω') are commensurate.

We can now state

Proposition 2 $\succsim$ satisfies A1-A8 iff it has a lexicarian representation $(\succsim^b, u)$ such that:

(i) There are $\hat{u} : C \rightarrow \mathbb{R}_+$, and, for each $\sim^b$ -equivalence class A , a probability vector $pr_A \in \Delta(\Omega)$ and $\delta_A > 0$ such that, for all $(c, t, \omega) \in A$,

$$u(c, t, \omega) = pr_A(\omega) \delta_A^t \hat{u}(c)$$

(ii) For every $c \in C$ and all $(t, \omega), (t', \omega') \in T \times \Omega$, $(c, t, \omega) \sim^b (c, t', \omega')$.

Moreover, in this case $\hat{u}$ is unique up to multiplication by a positive number, whereas pr_A and δ_A are unique if $\hat{u}$ is not identically zero.

Thus, retaining the weaker version of Archimedeanity (A4 rather than A4*), Proposition 2 allows for lexicarian preferences in the more elaborate model, reflecting dynamic and uncertainty considerations. However, axiom A8—clearly reflected in condition (ii) of the representation—restricts the lexicographic nature of preferences to the comparison of basic circumstances. Hence, other reasons for incommensurabilities, such as lexicographic probabilities as in Blume, Brandenburger, and Dekel (1991), are ruled out by A8.

Observe that the axioms, as stated, allow the probability and the discount factor to depend on the equivalence class. One may strengthen A5 and A6 to relate preferences across equivalence classes to obtain a tighter representation where pr_A and δ_A are independent of A .

4 Wants, Needs, and Positions

This section addresses the three interpretational questions raised above: (i) Should we insist on the sympathetic approach? (ii) Should the model reflect wants as well as needs? (iii) How should needs be described by positions?

4.1 Empathy or Sympathy?

The standard person-affecting approach to social welfare aggregates a vector of personalized utility functions, specifying the well-being of the various individuals. We are skeptical of such empathetic approaches, even with a fixed population and directly observable preferences. We use utilitarianism as a laboratory for exploring this issue, ending with a brief discussion of other aggregation rules.

First, if a planner is supposed to maximize a (weighted) sum of utilities, she has to choose for each individual a utility function, from a set of utility functions that represent the individual's preferences, but that are unique only up to an increasing affine transformation. The choice of a utility function from this set boils down to determining an individual's weight in the social aggregation. Asking individuals to provide their utility functions is tantamount to asking them how important they think they are. The result is likely to be an arms race: each agent faces unlimited incentives to multiply their utility function by ever-larger constants in an attempt to increase their allocation.

Harsanyi (1955) is silent on this question, devoting a careful argument to establishing that the planner must maximize some weighted sum of utilities while saying nothing about where the weights come from. Dhillon and Mertens (1999) resolve this issue by assuming that every individual's utility function has infimum zero and supremum one, and then invoking an anonymity axiom to assume that every individual receives the same weight in the summation. One problem with this 0-1 normalization is that individuals can differ in their concepts of "best" and "worst" outcomes. For example, a person of faith who believes in the afterlife can conceive of Heaven and Hell as possible outcomes, which might be beyond the scope of an atheist's mental space. As a result,

the atheist will have a higher weight in the social welfare function, required to normalize her mundane best- and worst-cases to the same scale as the believer's. Should the atheist promote maximization of such a sum, she could be thought of as telling the believer, "Well, you anyway care more about the afterlife, so why don't you give me all the material goods and live in poverty in the meantime?" This type of reasoning isn't completely foreign to human thought; indeed, believers who join monasteries might be viewed as following such trade-offs between earthly and heavenly payoffs. Yet, it seems problematic to endorse such a policy for a social planner, who would be penalizing the believer for her faith.¹⁵

Second, empathetic aggregation gives rise to counterintuitive implications even in the simplest of settings. Suppose there is a single private good x and three individuals, with utility functions

$$\begin{aligned} u_1(x_1) &= \sqrt{x_1} \\ u_2(x_2) &= x_2 \\ u_3(x_3) &= (x_3)^2. \end{aligned}$$

Let there be a total quantity X , and assume that we divide it among the three individuals so as to maximize the utilitarian criterion $\sum_{i=1}^3 \alpha_i u_i(x_i)$, for the case $\alpha_1 = \alpha_2 = \alpha_3$. Our focus is on the way the optimal allocation (x_1, x_2, x_3) varies as X varies. Figure 1 presents the qualitative solution. Never do all three people receive a positive quantity of resource, and there are both non-monotonicities and discontinuities of the allocations with respect to the total amount to be shared. Individual 2 initially receives nothing, then experiences a period in which all of any additional resource is given to her, and is then entirely deprived of the resource. Individual 3 initially receives nothing, then jumps to having the largest allocation, and eventually consumes the entire sup-

¹⁵One could instead suggest that the atheist can conceive of Heaven and Hell, but that he attaches zero probabilities to experiencing these outcomes after his death. But this raises the measurability problem again: it might be difficult to design a procedure that would tell us—and the atheist himself—what his utility should be for such extreme outcomes, given that he does not believe any act he takes would result in such outcomes with any positive probability.

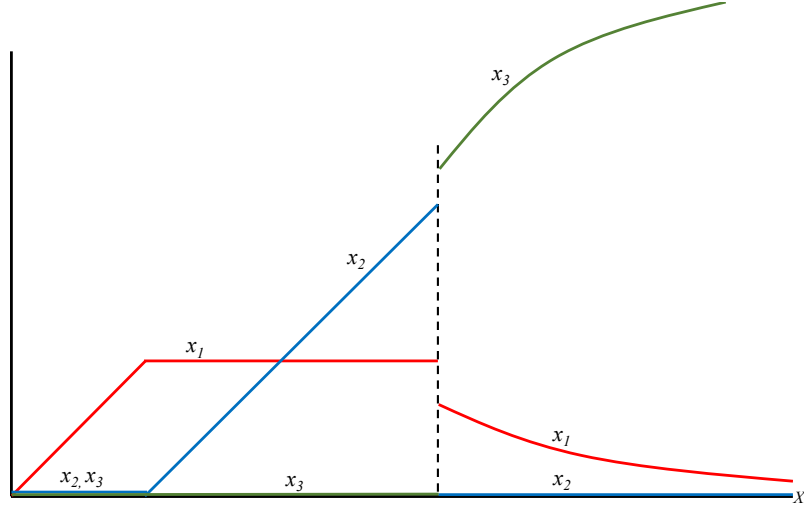


Figure 1: Quantity of resource allocated to person 1 (x_1 , red), person 2 (x_2 , blue when positive), and person 3 (x_3 , green when positive) as a function of total resource X .

ply. We might seek refuge from this outcome by attaching different weights to the people, but no assignment of weights will resolve all issues.

One may construct sufficiently many similarly bizarre examples to populate a zoo. Moreover, the counterintuitive conclusions do not depend on utilitarian aggregation. Suppose one adopted the Rawlsian order, leading the planner to equalize utility levels across individuals. Should individual 1 receive a meager allocation, simply because she is apparently more easily satisfied with smaller rations than either individuals 2 or 3? Should individual 3 receive the bulk of the resource, simply because he is morose and can be placated only by a massive allocation?

If we set out to avoid such quandaries, then the aggregation must not depend on the individual's utilities alone—the aggregation must not be purely empathetic. The planner must step in, at least supplementing the information contained in preferences with the planner's own view on how to aggregate the agents' well being. This brings us to the sympathetic approach.

4.2 Wants vs. Needs

Allowing the planner to invoke her own assessment of individuals' well-being raises challenges for the planner and raises the concern of the planner abusing her discretion. Both difficulties are mitigated by having the planner base her assessment of social welfare on needs rather than wants.

If the distinction between wants and needs is to make sense, we must be able to tell them apart. A cold beer on a hot summer day might be pleasurable, as well as solving the pain of thirst. A concert may be a pleasure to attend, and it may be solving a problem of boredom. In these cases, are we dealing with wants or needs? More generally, can there be any pleasure that doesn't satisfy a need of some sort?

We find an analogy between the want-need distinction and the question of interpersonal comparisons of utility. In both cases, a general, theoretically-neat resolution of the problem is elusive. In both, one might avoid these issues completely, rather than trod the marshes of shady psychological interpretations. However, doing so leaves not only the marshes but also the seas to politicians who fear no theoretical confusion. We believe there are sufficiently large bodies of clear water out there for economists to introduce these distinctions into their formal models. For instance, it appears evident that owning a private jet isn't a need in the sense that having dinner is. It seems equally obvious that, given the choice, a social planner should opt for feeding the hungry before supplying private jets to the well-nourished. Our preferred interpretation of the model reflects this distinction.

Can the planner form a reasonable idea of needs and wants? A positive answer is much more likely in the case of needs. The measurement of well-being by subjective self-reports of the extent to which wants are satisfied, used in psychological studies for decades, gave rise to Easterlin's Paradox (Easterlin 1973, 1974, Diener, 1984). The basic argument is that wants depend on one's aspirations and expectations, which change with experience as in adaptation level theory (Helson 1947, 1948). The implication is that, if we rely on self reports to determine well-being, it is doubtful that improving well-being is a

worthy or even feasible goal.¹⁶

Our view is that wants are irretrievably elusive while the planner can form a reasonable conception of individuals' needs. The need to focus on needs rather than wants becomes all the stronger when considering, for example, humanity 200 years hence. What will their wants be? Will they enjoy Beethoven more than techno music? Watching movies or playing computer games? Or are they more likely to enjoy pastimes we can't even conceive of? It seems hard to tell. By contrast, it doesn't seem to be far-fetched to assume that, like us, they will need food and shelter. Again, there is much less uncertainty (or ambiguity) about needs than about wants.

The function u appearing in (2), capturing the preferences of the planner, or the advocate motivating a social policy, or the moralist opining on the nature of a good society, may be informed by the preferences of individuals in society, but is not mechanically determined by them. The planner may recognize that all of the individuals in Section 4.1 value the good x , but may not respond to the fact that one person is easily satisfied with small quantities of x while another is satisfied only with large dollops of x , allowing the planner to escape the difficulties raised in Section 4.1.¹⁷

This sympathetic approach may appear to cede a great deal of arbitrary power to the planner. What would constrain her? What would prevent a government from imposing its own values on an entire population in an un-

¹⁶This point was made by Brickman, Coates, and Janoff-Bulman (1978) and Brickman and Campbell (1971), who suggested that there is no way to attain happiness apart from "stepping off the hedonic treadmill": if more resources raise one's aspirations, so that yet more resources are needed, the pursuit of happiness via material well-being is hopeless. This rather nihilistic view of well-being has been contested, and arguably the only clear conclusion is that research in the social sciences has not reached a conclusion about the definition and measurement of happiness. See Lucas, Dyrenforth, and Diener (2008), Veenhoven and Vergunst (2014), and Easterlin (2017) on Easterlin's paradox and Mancini, Bonanno, and Clark (2011) on the effect of major life events on well-being. See also Kauppinen (2013), Kahneman, Krueger, Schkade, Schwarz, and Stone (2004), Li, Jonah, Wong, and Chao (2019), Schwarz and Clore (1983), and Strack, Martin, and Schwarz (1988) for discussions of ascertaining well-being.

¹⁷Paul Bloom (2016) argued that sympathy is good for public policy, but empathy is not. A social planner can appropriately understand and make use of people's priorities without adopting their preferences as his own. Adam Smith (1759) and David Hume (1739-40, 1751) have similarly argued for limited empathy.

democratic way? We believe that the focus on needs, as opposed to wants, limits the risks of abuse of the social aggregation rule. To this end, the specification of needs should reflect a consensus within the society or polity to which the analysis applies. To allay fears that a reliance on consensus dooms us to incompleteness, we note that we do not seek consensus over social policies, but only the set of needs relevant for evaluating those policies. We would hope to reach a wide consensus over questions of the type, “Is food a basic need?” while expecting disagreement over “Should we tax the rich in order to feed the poor?” We envision a person advocating for a particular view of what constitutes a social good as appealing to our axioms in arguing that support for her argument is appropriately couched in terms of the lexitarian aggregation of needs. For support of her vision of what these needs are, she must then appeal to the consensus implicit in the structure and workings of the society around her.¹⁸

At a practical level, we may begin with some needs on which a consensus exists—say, food—and ask, which other needs are commensurate to these. For example, a person might be asked, Is education a need? We suggest that such a question be operationalized by the question, “Is there a number n such that depriving n children of education is worse than having one child being hungry?” Clearly, we do not purport to have any answers to such questions, and we don’t think it is the task of theorists to provide such answers. However, a theoretical model can render such a question more concrete, and help people decide what needs they would like to be included in the social planner’s considerations.

The interpretation of the model and the notion of needs clearly depend on the context. While our prime application is for policy decisions at the level of an economy, one can also think of social choice problems in other

¹⁸Views as to what constitutes a need and which things are desires that go beyond needs differ across societies and have evolved over time. Many European societies consider health care and education to be needs while this is not the case in the United States. Some societies consider that ability to participate in unhindered public speech a need; others do not. Most societies now take freedom from slavery and torture to be needs, though this is not quite universal. Internet access would recently have been unimaginable even as a want, while some now view it as a need. One might view the United Nations’ Universal Declaration of Human Rights as an attempt to make explicit the social consensus on needs.

contexts, where “needs” can be differently interpreted. Consider, for example, parents who have two children, Ada and Bobby. Ada loves mathematics, and obtained a PhD in the field and a tenured academic position. Bobby loves chess. He participates in chess clubs and tournaments, but supports himself by working in an unrelated and unrewarding job. The parents may well consider self-fulfillment as a need, and provide more financial support to Bobby than to Ada, so that Bobby can satisfy the self-fulfillment need as does Ada. In contrast, self-fulfillment may not qualify as a need the social planner takes into account when devising the income tax code.

4.3 Modeling Needs by Positions

We use the notion of a position to describe the extent to which needs are satisfied. Suppose that there are R needs, and that for each person the extent to which need $r \in \{1, \dots, R\}$ is met can be described by a number drawn from a finite set N_r . How should positions describe the degrees to which these needs are satisfied?

Let us first analyze a simple example. Suppose that there are two needs, food and space trips, with the former deemed incommensurably more basic than the latter. Each need is classified as unsatisfied or wholly satisfied, so that $N_1 = N_2 = \{0, 1\}$. How should we define P ? It seems natural to think of a position as an element of $N_1 \times N_2$, so that, knowing that a person is at a position $p \in N_1 \times N_2$, we know everything about her well-being. With $P = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$, a profile I would indicate that there are $I(0, 0)$ people who are starving and make no space trips, $I(1, 0)$ people who have enough to eat but don’t travel in space, and so on. A lexitarian model $(\succsim^b, u)$ with a trivial $\succsim^b$ (i.e., a model in which we have $\succsim^b = P \times P$ and the strong Archimedean axiom holds) would face the social planner with two choices: either u ignores the second coordinate of the positions, or else one is driven to the conclusion that sufficiently many space trips for some individuals compensate for the starvation of others. To avoid this conclusion, one may wish to use the additional freedom of the lexitarian model. Suppose that $(1, 0) \sim^b (1, 1) \succ^b (0, 0) \sim^b (0, 1)$. That is, the relation $\succsim^b$ has two equivalence

classes, one in which the individual is well-fed, another in which she starves, with the former ranked strictly above the latter. Within each indifference class, the function u may still represent a preference for space travel, so that we can have $u(1, 1) > u(1, 0)$ and $u(0, 1) > u(0, 0)$. However, in this case there will exist a number $n \geq 1$ such that $n(u(1, 1) - u(1, 0)) > u(1, 0)$, meaning that providing space travel to n well-fed individuals can compensate for moving one well-fed individual into the starving $\succsim^b$ equivalence class.

Our focus on needs suggests a different modeling choice: defining P as the union of N_1, N_2 , rather than their cross product. Explicitly, we can have $P = \{nf, f, ns, s\}$ where “ f ” and “ nf ” stand for “food” and “no-food”, respectively, and “ s ” and “ ns ” are similarly defined for space travel. With a trivial $\succsim^b$ one faces the same problem as above: either $u(ns) = u(s)$, and the social planner ignores preferences for space trips, or else there is a number of space trips that would justify starvation. However, if we have $f \sim^b nf \succ^b s \sim^b ns$, and $u(f) > u(nf)$, $u(s) > u(ns)$, then we obtain a ranking that first considers food, and only when the number of hungry individuals is identical across two profiles, does it compare space trips. This allows the planner to recognize that food and space trips both have value, but to reject trade-offs between the two.

In such a model, each individual is counted exactly once among the positions of a given need. That is, a profile I should satisfy $I(nf) + I(f) = I(ns) + I(s)$ and these equal the total number of individuals in society. However, our model assumes that preferences are defined also for other I ’s, whose sum across different needs is not constant. Moreover, the definition of commensurability involves indicator profiles (the profiles $\mathbf{1}_p, \mathbf{1}_q$ for positions p, q) having 1 at a given position and 0 elsewhere. What type of society do these represent? And will we want to say that $\mathbf{1}_{nf} \succ \mathbf{1}_s$, as required by the definition of $nf \succ^b s$?

In order to cope with this problem, we propose to omit from the model the lowest level of need satisfaction for each need. In this example, we define only two positions: $P = \{f, s\}$. A profile I now suggests that $I(f)$ individuals are well-fed (and the others have no food), and $I(s)$ enjoy space trips (while the others do not). In this formulation, every profile I has a clear interpretation

and it makes sense to consider indicator profiles. Specifically, we will likely have $\mathbf{1}_f \succ \mathbf{1}_s$: a society with one well-fed individual who does not enjoy space trips is preferred to a society with a starving space traveler. Further, we can have $f \succ^b s$ if no number of space trips can compensate for a single person's starvation.

More generally, with R needs, we can define $N'_r = N_r \setminus \{\min N_r\}$ and then consider the union of these as the set of positions: $P = \bigcup_{r=1}^R N'_r$. The planner's preferences over profiles would reveal which needs are considered commensurably to which. For instance, if we add the needs for medication and for opera to the example above, say, $N_3 = N_4 = \{0, 1\}$, the model will have four positions $P = \{f, s, m, o\}$ (where “ m ” stands for “having medication” and “ o ” – for opera”). Any profile $I : P \rightarrow \mathbb{Z}_+$ has a natural interpretation, and preferences over these might indicate that $f \sim^b m \succ^b s \sim^b o$, so that food and medication belong to the same equivalence class, and between them tradeoff is conducted in the utilitarian fashion. Similarly, space trips and operas are in an equivalence class, and tradeoffs between them are also dictated by utilitarian addition. But the former are incommensurably more basic than the latter.

Notice that in this model, the social planner will be indifferent to the addition of people whose needs are all at the minimal level. That is, if we add individuals who, for each r , have the r -th need at $\min N_r$, the profile that represents this population is unchanged. Thus, in this model there is no life strictly not worth living, but the very worst life is equivalent to not-living. This interpretation can be used to define the sets N_r : when thinking of a need, we may introduce into the set the lowest conceivable level above which there's some point in living.

Importantly, modeling positions as the union, rather than the cross product, of $\{N_r\}_r$ implies that the social planner thinks of needs in a separably additive way. The additive structure of the model (owing mostly to the Union Axiom) means that the preference one has over profiles are independent across needs. In a sense, in this formulation of “positions”, needs are the carriers of the planner's utility, while individuals are collections of needs. It is as if a social planner who observes a society of m individuals sees m stomachs that

need to be filled, m bodies that need to be sheltered, m minds that need to be entertained, and so forth. This separability would typically be satisfied by government policies, if only for administrative reasons: we do not expect a government agency to deprive an individual of public housing because they already received free meals. However, putting administrative difficulties aside, we find that such a separability is also desirable from a normative point of view (for instance because it is consistent with the general principle of incommensurability in needs).

5 Discussion: Common Practice and Government Budgets

The lexitarian model captures several features of the common practice of government policy. First, economic policy goes beyond Pareto optimality, and, in practice, makes interpersonal comparisons of utility despite the official position of microeconomic textbooks. For example, progressive taxation is typically motivated by the intuition that it is easier for the rich than for the poor to part with a given amount of money. Second, choices are made in ways that seem to be intuitively in line with the need/want distinction. Governments allocate budgets to cope with hunger and disease, but not for space trips as a recreational activity.

Beyond these extremes, people and polities have different preferences. For instance, are cultural activities a need that is on par with food and medication? Should the government finance, say, a state opera? Or, to make the question more concrete, is one child's hunger worth n people enjoying an opera? Our model is designed to deal with such problems, allowing one to say that space tourism isn't considered a "need" at all, while opera might be a need which is in a lower category than food.

Actual government decisions necessarily make trade-offs between all of the programs the government funds. Should a government allocate some of its budget to a state opera, it invariably provides less funds to food programs and to life-saving medical treatments. It is fair to ask, therefore, what is the use of

a lexicarian model: in the final analysis, are there any policy decisions in which one finds equality at levels $(u_1, \dots, u_\ell)$ and hence the decision is determined by differences in the function $u_{\ell+1}$?

Our answer is two-fold. First, as is the case with any model, our model involves various idealizations. While perfect equality of $(u_1, \dots, u_\ell)$ is indeed unlikely, in reality our ability to measure degrees of need-satisfaction is limited, and utilities can be measured up to just-noticeable-differences. A government may decide to allocate some funds to the opera, considering amounts that would not make a noticeable difference in the degree of satisfaction of more basic needs of the population at large. Second, we imagine a procedure by which a government is expected to justify policy decisions by exhibiting a model that rationalizes them. Faced with a proposal to allocate funds to food assistance, the government might identify a list of priority classes reflecting the levels in our ordering $\succsim^b$ and a description of the trade-offs within classes under which the policy constitutes an improvement.¹⁹ Importantly, various feasibility constraints may render it impossible to satisfy more highly-regarded needs, whereas less crucial ones can be satisfied to higher degrees. The lexicarian model provides the government with a convenient language for expressing these priorities and trade-offs.

6 Proofs

The proof of Theorem 1 builds on the proof of Theorem 2, and so we present the latter first.

6.1 Proof of Theorem 2

It is straightforward that the representation (2) implies that $\succsim$ satisfies A1-A3. We thus turn to prove sufficiency of the axioms, from which the uniqueness of the representation will also follow. Naturally, the idea is to extend the relation, given for natural-number vectors, to real-valued vectors and apply

¹⁹Gilboa and Samuelson (2022) offer a similar criterion for an agent, such as a portfolio manager or a government, making decisions under uncertainty on behalf of others.

a separation theorem. More specifically, we wish to successively extend $\succsim$ to vectors of integer and rational numbers, and then apply a separation theorem to the convex hull of the rational-valued vectors. To this end we define the following sets of vectors:

$$\begin{aligned}\mathcal{I}_{\mathbb{Z}} &= \{I : P \rightarrow \mathbb{Z} \mid \#supp(I) < \infty\} \\ \mathcal{I}_{\mathbb{Q}} &= \{I : P \rightarrow \mathbb{Q} \mid \#supp(I) < \infty\}\end{aligned}$$

For two sets $A \subset B$ and two binary relations on them, $\succsim \subset A \times A$ and $\succsim' \subset B \times B$, we say that $\succsim'$ is an *extension* of $\succsim$ if it extends both the weak and strict preference, that is, if, $\succsim \subset \succsim'$ and $\succ \subset \succ'$. We will also need to refer to the versions of axioms A1-A3 where the vectors are in $\mathcal{I}_{\mathbb{Z}}$ and in $\mathcal{I}_{\mathbb{Q}}$ (rather than only in $\mathcal{I}$). To simplify notation, we will use the same terms “A1”, “A2”, “A3”, rather than redefine them for these domains.

Assume, then, that $\succsim \subset \mathcal{I} \times \mathcal{I}$ satisfies A1-A3. We first extend A2 to apply also for vectors K that are integer-valued but not necessarily nonnegative.

Lemma 1 *There exists a unique $\succsim_{\mathbb{Z}}$ on $\mathcal{I}_{\mathbb{Z}}$ satisfying A1-A3 and extending $\succsim$ (from $\mathcal{I}$ to $\mathcal{I}_{\mathbb{Z}}$).*

Proof: Let there be given $I, J \in \mathcal{I}_{\mathbb{Z}}$. Define $I \succsim_{\mathbb{Z}} J$ if there exists $K \in \mathcal{I}$ such that $I + K \succsim J + K$ (here and in the sequel, the claim that $\succsim$ holds between two vectors is taken to mean that they are both in $\mathcal{I}$ and that the relation holds between them). Clearly, $\succsim_{\mathbb{Z}}$ extends $\succsim$. Completeness of $\succsim$ on $\mathcal{I}$ implies completeness of $\succsim_{\mathbb{Z}}$ on $\mathcal{I}_{\mathbb{Z}}$. To see that $\succsim_{\mathbb{Z}}$ is transitive, assume that for $I, J, L \in \mathcal{I}_{\mathbb{Z}}$ we have $I \succsim_{\mathbb{Z}} J$ and $J \succsim_{\mathbb{Z}} L$. Let $K, K' \in \mathcal{I}$ be such that (i) $I + K \succsim J + K$ and (ii) $J + K' \succsim L + K'$. Applying A2 to (i) and K' and then to (ii) and K , we have $I + K + K' \succsim J + K + K' \succsim L + K + K'$ and thus $I \succsim_{\mathbb{Z}} L$.

Next we wish to show that $\succsim_{\mathbb{Z}}$ satisfies A2 on $\mathcal{I}_{\mathbb{Z}}$. Let there be given $I, J, K \in \mathcal{I}_{\mathbb{Z}}$, with $I \succsim_{\mathbb{Z}} J$. This means that there exists $K' \in \mathcal{I}$ such that $I + K' \succsim J + K'$. Assume first that $K \in \mathcal{I}$. By A2 $I + K' + K \succsim J + K' + K$ so that $I + K \succsim_{\mathbb{Z}} J + K$. Similarly, $I + K \succsim_{\mathbb{Z}} J + K$ implies $I \succsim_{\mathbb{Z}} J$ (using $(K + K') \in \mathcal{I}$).

For a general (not necessarily nonnegative) $K \in \mathcal{I}_{\mathbb{Z}}$, let $K^+, K^- \in \mathcal{I}$ be such that $K = K^+ - K^-$. Then $I + K = I + K^+ - K^-$. Denote $I' = I - K^- \in \mathcal{I}_{\mathbb{Z}}$. Similarly, let $J' = J - K^-$. By the restricted A2 (applied to the nonnegative $K^- \in \mathcal{I}$), $I' \lesssim_{\mathbb{Z}} J'$, iff $I' + K^- \lesssim_{\mathbb{Z}} J' + K^-$, that is iff $I \lesssim_{\mathbb{Z}} J$. Applying the restricted A2 again to I', J' , this time with the non-negative $K^+ \in \mathcal{I}$, we get $I' \lesssim_{\mathbb{Z}} J'$ iff $I' + K^+ \lesssim_{\mathbb{Z}} J' + K^+$. Thus, $I \lesssim_{\mathbb{Z}} J$ iff $I' + K^+ \lesssim_{\mathbb{Z}} J' + K^+$. It is left to note that $I' + K^+ = I - K^- + K^+ = I + K$ and similarly $J' + K^+ = J + K$, so that $I \lesssim_{\mathbb{Z}} J$ iff $I + K \lesssim_{\mathbb{Z}} J + K$.

We now turn to A3. Assume that $I, J, K, L \in \mathcal{I}_{\mathbb{Z}}$, and $K \succ_{\mathbb{Z}} L$. Let $K' \in \mathcal{I}$ be such that $I + K', J + K' \in \mathcal{I}$. As $K \succ_{\mathbb{Z}} L$, there exists $K'' \in \mathcal{I}$ such that $K + K'' \succ L + K''$. Applying A3 (for $\succ$ on $\mathcal{I}$) to $I + K', J + K'$ and the pair $K + K'' \succ L + K''$, we conclude that for some $n \geq 0$ we have $I + K' + n(K + K'') \succ J + K' + n(L + K'')$. This implies $I + nK + (K' + nK'') \succ_{\mathbb{Z}} J + nL + (K' + nK'')$ and, by A2, this is equivalent to $I + nK \succ_{\mathbb{Z}} J + nL$.

By similar arguments, $\lesssim_{\mathbb{Z}}$ is the unique extension of $\lesssim$ to $\mathcal{I}_{\mathbb{Z}}$ that satisfies the three axioms. $\square$

Lemma 2 *For all $I, J \in \mathcal{I}_{\mathbb{Z}}$, (i) $I \lesssim_{\mathbb{Z}} J$ iff $I - J \lesssim_{\mathbb{Z}} 0$; (ii) for all $n > 1$, $I \lesssim_{\mathbb{Z}} J$ iff $nI \lesssim_{\mathbb{Z}} nJ$.*

Proof: Follows from A2 and transitivity. $\square$

We can now make the next step into the rationals.

Lemma 3 *There exists a unique $\lesssim_{\mathbb{Q}}$ on $\mathcal{I}_{\mathbb{Q}}$ satisfying A1-A3 and extending $\lesssim_{\mathbb{Z}}$ (from $\mathcal{I}_{\mathbb{Z}}$ to $\mathcal{I}_{\mathbb{Q}}$).*

Proof: Let there be given $I, J \in \mathcal{I}_{\mathbb{Q}}$. Define $I \lesssim_{\mathbb{Q}} J$ if there exists $n \geq 1$ such that $nI \lesssim_{\mathbb{Z}} nJ$. Clearly, $\lesssim_{\mathbb{Q}}$ extends $\lesssim_{\mathbb{Z}}$. Completeness of $\lesssim_{\mathbb{Q}}$ follows from completeness of $\lesssim_{\mathbb{Z}}$ on $\mathcal{I}_{\mathbb{Z}}$. To see that transitivity holds as well, assume that $I, J, K \in \mathcal{I}_{\mathbb{Q}}$ with $I \lesssim_{\mathbb{Q}} J$ and $J \lesssim_{\mathbb{Q}} K$. Let n be such that $nI \lesssim_{\mathbb{Z}} nJ$ and let m be such that $mJ \lesssim_{\mathbb{Z}} mK$. Applying Lemma 2 to both, we conclude that $nmI \lesssim_{\mathbb{Z}} nmJ \lesssim_{\mathbb{Z}} nmK$ transitivity of $\lesssim_{\mathbb{Z}}$ implies that of $\lesssim_{\mathbb{Q}}$.

Next consider A2. It suffices to show that $I \lesssim_{\mathbb{Q}} J$ implies $I + K \lesssim_{\mathbb{Q}} J + K$ (as the converse argument is symmetric). Let there be given $I, J, K \in \mathcal{I}_{\mathbb{Q}}$,

with $I \succsim_{\mathbb{Q}} J$. Let $n \geq 1$ be such that $nI \succsim_{\mathbb{Z}} nJ$. Let $m \geq 1$ satisfy $mK \in \mathcal{I}_{\mathbb{Z}}$. By Lemma 2, $nmI \succsim_{\mathbb{Z}} nmJ$ and A2 yields $nmI + nmK \succsim_{\mathbb{Z}} nmJ + nmK$, that is, $I + K \succsim_{\mathbb{Q}} J + K$.

We now turn to A3. Assume that $I, J, K, L \in \mathcal{I}_{\mathbb{Q}}$, and $K \succ_{\mathbb{Q}} L$. Let $m \geq 1$ satisfy $mJ, mI \in \mathcal{I}_{\mathbb{Z}}$. For some $m' \geq 1$, $m'K \succ_{\mathbb{Z}} m'L$. By Lemma 2, $mm'K \succ_{\mathbb{Z}} mm'L$. Applying A3 (on $\succsim_{\mathbb{Z}}$ on $\mathcal{I}_{\mathbb{Z}}$) to $mm'I, mm'J, mm'K, mm'L$ we infer that there exists $n \geq 0$ such that $mm'I + nmm'K \succ_{\mathbb{Z}} mm'J + nmm'L$, which, by definition, means that $I + nK \succ_{\mathbb{Q}} J + nL$.

Finally, similar arguments imply that the extension $\succsim_{\mathbb{Q}}$ is unique. $\square$

Lemma 4 *For all $I, J \in \mathcal{I}_{\mathbb{Q}}$, (i) $I \succsim_{\mathbb{Q}} J \iff I - J \succsim_{\mathbb{Q}} 0$; (ii) for all $n > 1$, $I \succsim_{\mathbb{Q}} J \iff nI \succsim_{\mathbb{Q}} nJ$.*

Proof: Identical to the proof of Lemma 2. $\square$

It is worth mentioning explicitly the following:

Lemma 5 *For all $I, J \in \mathcal{I}_{\mathbb{Q}}$, and every rational number $\alpha > 0$, $I \succsim_{\mathbb{Q}} J$ iff $\alpha I \succsim_{\mathbb{Q}} \alpha J$ (and thus $I \succ_{\mathbb{Q}} J$ iff $\alpha I \succ_{\mathbb{Q}} \alpha J$).*

Proof: Follows from Lemma 4. $\square$

We turn to the definition of the function u . If $I \sim J$ for all $I, J \in \mathcal{I}$, select $u(\cdot) \equiv 0$. Otherwise, assume that $I^* \succ J^*$ for some $I^*, J^* \in \mathcal{I}$. Consider a finite $P_0 \subset P$ such that $P^* \equiv \text{supp}(I^*) \cup \text{supp}(J^*) \subset P_0$. We will define u for such a subset P_0 and then show how these definitions can be “patched” together to a definition of u on all of P . Let $\mathcal{I}_{\mathbb{Q}_0}$ be the set of vectors in $\mathcal{I}_{\mathbb{Q}}$ that vanish outside of P_0 . Define

$$\begin{aligned} A &= \{I \in \mathcal{I}_{\mathbb{Q}_0} \mid I \succ_{\mathbb{Q}} 0\} \\ B &= \{I \in \mathcal{I}_{\mathbb{Q}_0} \mid 0 \succ_{\mathbb{Q}} I\} \\ E &= \{I \in \mathcal{I}_{\mathbb{Q}_0} \mid I \sim_{\mathbb{Q}} 0\} \end{aligned}$$

We consider A, B, E as subsets of the finite-dimensional Euclidean space $\mathbb{R}^{|P_0|}$, and observe the following: (I) A, B, E are pairwise disjoint; (II) $A \cup B \cup E = \mathcal{I}_{\mathbb{Q}_0}$; (III) $A = -B$; (IV) $0 \in E$; (V) for all $I, J \in \mathcal{I}_{\mathbb{Q}_0}$, $I \succ_{\mathbb{Q}} J \iff$

$I - J \in A, J \succ_{\mathbb{Q}} I \iff I - J \in B$, and $I \sim_{\mathbb{Q}} J \iff I - J \in E$; and (VI) $A, B \neq \emptyset$.

Lemma 6 *Suppose that $I, J \in A(B, E)$ and $\alpha \in [0, 1] \cap \mathbb{Q}$. Then $\alpha I + (1 - \alpha) J \in A(B, E)$.*

Proof: Follows from Lemma 5, A2, and transitivity. $\square$

In the following, for $I \in \mathbb{R}^{|P_0|}$ and $\varepsilon > 0$, $N_\varepsilon(I)$ denotes the ε -neighborhood of I in $\mathbb{R}^{|P_0|}$.

Lemma 7 *Let $I \in A(B)$. There exists $\varepsilon > 0$ such that for every $J \in N_\varepsilon(I) \cap \mathcal{I}_{\mathbb{Q}_0}$, we have $J \in A(B)$.*

Proof: Let there be given $I \in A$. For $1 \leq j \leq |P_0|$, let e_j be the j -th unit vector in $\mathbb{R}^{|P_0|}$. We wish to find an $\varepsilon_j > 0$ such that $[I - \varepsilon_j e_j, I + \varepsilon_j e_j] \cap \mathcal{I}_{\mathbb{Q}_0} \subset A$. Consider $K^+ = I + e_j \in \mathcal{I}_{\mathbb{Q}_0}$. By A3 there exists $n \geq 1$ such that $K^+ + nI \succ_{\mathbb{Q}} 0$. Hence by Lemma 4), $\frac{1}{n+1} [K^+ + nI] \succ_{\mathbb{Q}} 0$. Thus, for $\varepsilon_j^+ = \frac{1}{n+1}$ we have $[I, I + \varepsilon_j^+ e_j] \cap \mathcal{I}_{\mathbb{Q}_0} \subset A$. Similarly, there exists $\varepsilon_j^- > 0$ such that $[I - \varepsilon_j^- e_j, I] \cap \mathcal{I}_{\mathbb{Q}_0} \subset A$ —and we define $\varepsilon_j = \min(\varepsilon_j^+, \varepsilon_j^-)$.

Let $\bar{\varepsilon} = \min_j \varepsilon_j$ and set $\varepsilon = \frac{\bar{\varepsilon}}{\sqrt{|P_0|}} > 0$. It follows that the ε -ball around I is included in the convex hull of $\{I - \bar{\varepsilon} e_j, I + \bar{\varepsilon} e_j\}_{j \leq |P_0|}$. Consider a rational point $J \in N_\varepsilon(I) \cap \mathcal{I}_{\mathbb{Q}_0}$. By Carathodory's Theorem, there are $l \leq |P_0| + 1$ points $\{I_r\}_{r \leq l}$ in $\{I - \bar{\varepsilon} e_j, I + \bar{\varepsilon} e_j\}_{j \leq |P_0|}$ and nonnegative numbers $\{\alpha_r\}_{r \leq l}$ adding up to 1 such that $J = \sum_{r \leq l} \alpha_r I_r$, and, furthermore, these $\{\alpha_r\}_{r \leq l}$ are unique (for the given set $\{I_r\}_{r \leq l}$). This implies that $\{\alpha_r\}_{r \leq l}$ are rational (as the unique solution to a system of equations with rational numbers). It follows, from iterated applications of Lemma 6, that $J = \sum_{r \leq l} \alpha_r I_r \in A$.

The argument for B is symmetric. $\square$

Lemma 8 *For every $I \in E$ there are arbitrarily close points $J^+ \in A$ and $J^- \in B$.*

Proof: Follows from Lemma 4, A2, and transitivity. $\square$

Lemma 9 *There exists $u : P_0 \rightarrow \mathbb{R}$ such that, for all $I, J \in \mathcal{I}_{\mathbb{Q}_0}$*

$$\begin{aligned}
I &\succsim_{\mathbb{Q}_0} J \\
&\text{iff} \\
\sum_{p \in P_0} I(p) u(p) &\geq \sum_{p \in P_0} J(p) u(p)
\end{aligned}$$

Further, u is unique up to multiplication by a positive number.

Proof: We know that A and B are closed under rational convex combinations. Consider $A^0 = \text{int}(\text{conv}(A))$ and $B^0 = \text{int}(\text{conv}(B))$, where conv refers to the convex hull in $\mathbb{R}^{|P_0|}$ (including not-necessarily rational points, and not-necessarily rational convex combinations). By Lemma 7 the sets $A^0, B^0 \subset \mathbb{R}^{|P_0|}$, which are open and convex, are non-empty (and full-dimensional). By Lemma 7 we also know that $A \subset A^0$ and $B \subset B^0$. Moreover, for any $I \in A^0(B^0)$ there exists $\varepsilon > 0$ such that $N_\varepsilon(I) \cap \mathcal{I}_{\mathbb{Q}_0} \subset A(B)$. In particular, this means that $A^0 \cap B^0 = \emptyset$: if there were $x \in A^0 \cap B^0$, we would be able to find two neighborhoods thereof, such that in one all rational points have to be in A and in the other—in B , in contradiction to $A \cap B = \emptyset$.

Thus we can apply a separating hyperplane theorem to conclude that there exists $u : P_0 \rightarrow \mathbb{R}$ and a number $c \in \mathbb{R}$ such that

$$I \in A^0 \implies I \cdot u > c \quad (\text{i})$$

$$I \in B^0 \implies I \cdot u < c \quad (\text{ii})$$

We first note that the implications in (i) and in (ii) can also be reversed. To see this, assume that $I \cdot u > c$. Choose $\varepsilon > 0$ such that for every $J \in N_\varepsilon(I)$, we have $J \cdot u > c$. We argue that, if $J \in \mathcal{I}_{\mathbb{Q}_0}$, we have $J \in A$. Indeed, if $J \in B$ then $J \in B^0$ and (ii) would imply $J \cdot u < c$. If, however, $J \in E$, then, by Lemma 8, there are points $J' \in B$ arbitrarily close to J , so that we can pick $J' \in N_\varepsilon(I) \cap B$. This would contradict (ii) again. Thus all rational points in $N_\varepsilon(I)$ are in A and $I \in A^0$ follows. The argument is symmetric for B (and the converse of (ii)).

Next, observe that for $I \in E$ we have to have $I \cdot u = c$, because an inequality would imply that I is in A or in B . Finally we observe that $c = 0$ because $0 \in E$.

We conclude that for all $I \in \mathcal{I}_{\mathbb{Q}}$

$$\begin{aligned} I \in A &\iff I \cdot u > 0 \\ I \in E &\iff I \cdot u = 0 \\ I \in B &\iff I \cdot u < 0 \end{aligned}$$

which means that, for all $I, J \in \mathcal{I}_{\mathbb{Q}_0}$, $I \succsim_{\mathbb{Q}_0} J$ ($I \succ_{\mathbb{Q}_0} J$) iff $I - J \in A \cup E$ ($I - J \in A$) iff $(I - J) \cdot u \geq 0$ (> 0)—and u is unique up to multiplication by a positive constant $\lambda > 0$. $\square$

To conclude the proof of the theorem, we now define u for all p . We apply Lemma 9 and fix one $u_0 : P_0 \rightarrow \mathbb{R}$ that represents $\succsim_{\mathbb{Q}_0}$ on $\mathcal{I}_{\mathbb{Q}_0}$. For every $p \notin P_0$, we consider $P_p = P \cup \{p\}$ and the corresponding $\mathcal{I}_{\mathbb{Q}_p}$. There exists a unique function u_p that extends u_0 to P_p and represents $\succsim_{\mathbb{Q}_p}$. Let $u(p) = u_p(p)$. Thus u is defined for all of P .

To see that u so defined represents $\succsim_{\mathbb{Q}}$ on all of $\mathcal{I}_{\mathbb{Q}}$, consider $K, L \in \mathcal{I}_{\mathbb{Q}}$. Limit attention to $P' = P^* \cup \text{supp}(K) \cup \text{supp}(L)$. By the same arguments, there exists a $u' : P' \rightarrow \mathbb{R}$ that represents $\succsim_{\mathbb{Q}'}$. It thus also represents $\succsim_{\mathbb{Q}_0}$ on $\mathcal{I}_{\mathbb{Q}_0}$ and $\succsim_{\mathbb{Q}_p}$ on $\mathcal{I}_{\mathbb{Q}_p}$ for every $p \in P'$. Hence it has to equal u up to multiplication by a positive constant. $\square$

6.2 Proof of Theorem 1

We begin with sufficiency of the axioms. We first note that

Lemma 10 *For all $I, J, K, L \in \mathcal{I}$ and all $m \geq 1$,*

- (i) $I \succsim J$ and $K \succ (\succ) L$ imply $I + K \succ (\succ) J + L$;
- (ii) If $K \sim L$ then $I \succsim J$ iff $I + K \succsim J + L$;
- (iii) $I \succ (\succ) J$ implies $mI \succ (\succ) mJ$.

Proof: (i) follows from two applications of A2 and transitivity: $I \succsim J$ iff $I + K \succsim J + K$ and $K \succ (\succ) L$ iff $J + K \succ (\succ) J + L$.

(ii) follows from two applications of (i) if $I \succsim J$ then $K \succ (\succ) L$ implies $I + K \succ (\succ) J + L$. Conversely, if $I + K \succ (\succ) J + L$, had $J \succ I$ been the case, (i) would yield $J + L \succ I + K$, a contradiction.

(iii) follows from iterated applications of (i). $\square$

We now turn to define the relation $\succsim$: For $p, q \in P$, $p \succsim \cdot q$ if $\exists k \geq 1$ such that $k\mathbf{1}_p \succsim \mathbf{1}_q$.

Lemma 11 (i) $\succsim \cdot$ is complete;

(ii) $\succsim \cdot$ is transitive;

(iii) $\sim \cdot$ coincides with commensurability.

Proof: (i) follows from completeness of $\succsim$, say, applied to $\mathbf{1}_p$ and $\mathbf{1}_q$.

(ii) follows from transitivity of $\succsim$ and Lemma 10: assume that $p \succsim \cdot q$ and $q \succsim \cdot r$. Let $k, l \geq 1$ be such that $k\mathbf{1}_p \succsim \mathbf{1}_q$ and $l\mathbf{1}_q \succsim \mathbf{1}_r$. Part (iii) of Lemma 10 implies that $kl\mathbf{1}_p \succsim l\mathbf{1}_q$ and transitivity—that $kl\mathbf{1}_p \succsim \mathbf{1}_r$, hence $p \succsim \cdot r$.

Finally, (iii) is immediate. $\square$

Let us define the $\sim$ -equivalence classes to be $\mathcal{A} \equiv P / \sim \cdot$. We use $\succsim \cdot$ over $\mathcal{A}$ with the obvious meaning. We can write, for each I ,

$$I = \sum_{A \in \mathcal{A}} I_A$$

and note that $I_A = 0$ for all but finitely many A 's. Equivalently, for every I there is a finite set of $\sim$ -equivalence classes, $(A_1, \dots, A_m)$ such that $A_i \succ \cdot A_{i+1}$ (for $1 \leq i < m$) and

$$I = \sum_{i \leq m} I_{A_i}$$

(and $I_B = 0$ for all $B \in \mathcal{A} \setminus \{A_1, \dots, A_m\}$).

The key step in the proof is to show that, if two profiles I, J can be ranked on a given $A \in \mathcal{A}$, anything that happens in lower indifference classes of positions does not matter. Formally,

Lemma 12 Let there be given $I, J \in \mathcal{I}$ and $A \in \mathcal{A}$ such that

(i) $I_A \succ J_A$

(ii) $I_B \sim J_B$ for all $B \in \mathcal{A}$ with $B \succ \cdot A$;

Then $I \succ J$.

Proof: We first write

$$\begin{aligned} I &= \sum_{\substack{B \in \mathcal{A} \\ B \succ \cdot A}} I_B + I_A + \sum_{\substack{C \in \mathcal{A} \\ A \succ \cdot C}} I_C \\ J &= \sum_{\substack{B \in \mathcal{A} \\ B \succ \cdot A}} J_B + J_A + \sum_{\substack{C \in \mathcal{A} \\ A \succ \cdot C}} J_C \end{aligned}$$

and it will be useful to look also at

$$\begin{aligned} I^* &= \sum_{\substack{B \in \mathcal{A} \\ B \succ \cdot A}} I_B + I_A \\ J^* &= \sum_{\substack{B \in \mathcal{A} \\ B \succ \cdot A}} J_B + J_A \end{aligned}$$

Because $I_A \succ J_A$, we can Lemma 10 consecutively for every $B \succ \cdot A$ such that $I_B \neq 0$ or $J_B \neq 0$ (and there are finitely many such B 's) to conclude that $I^* \succ J^*$. We now wish to prove, that we can change I^* and J^* on any position p in C with $A \succ \cdot C$, without changing the strict preference. As there are finitely many positions that distinguish (I, J) from (I^*, J^*) , the argument can be used inductively to show that $I^* \succ J^*$ does indeed imply $I \succ J$.

Let there be given such a p , so that $q \succ \cdot p$ for every $q \in A$. Consider the modified profiles

$$\begin{aligned} I_p^* &= I^* + I(p) \mathbf{1}_p \\ J_p^* &= J^* + J(p) \mathbf{1}_p \end{aligned}$$

We wish to show that $I_p^* \succ J_p^*$. We will use A4 to conclude that $I_p^* \succsim I^*$, and thus it would suffice to prove that $I^* \succ J_p^*$.²⁰ Assume not. Then

$$J_p^* = J^* + J(p) \mathbf{1}_p \succsim I^*$$

Consider any $q \in A$. Apply the Archimedean axioms to the four profiles $0, \mathbf{1}_q, I_A, J_A$ whose supports are all included in A . Since $I_A \succ J_A$, there exists $k \geq 1$ such that $(0+) kI_A \succ \mathbf{1}_q + kJ_A$. (That is, adding one at position q does

²⁰If A4 is not assumed, the argument is similar.

not suffice to reverse the strict preference $kI_A \succ kJ_A$.) However, $J^* + J(p) \mathbf{1}_p \succsim I^*$ implies (by Lemma 10 (iii)) that

$$kJ^* + kJ(p) \mathbf{1}_p \succsim kI^*$$

Using Lemma 10 (ii), when applied to all B with $B \succ \cdot A$ (for which $I_B \sim J_B$ and $kI_B \sim kJ_B$), we also have

$$kJ_A + kJ(p) \mathbf{1}_p \succsim kI_A \succ \mathbf{1}_q + kJ_A$$

The preference $kJ_A + kJ(p) \mathbf{1}_p \succ \mathbf{1}_q + kJ_A$ implies, using Lemma 10 (i), $kJ(p) \mathbf{1}_p \succ \mathbf{1}_q$ which means that $p \succsim \cdot q$ —a contradiction.

Thus we have $I_p^* \succ J_p^*$ and, by repeating the argument inductively for every p in $\text{supp}(I) \cup \text{supp}(J)$ with $q \succ \cdot p$ for $q \in A$, we conclude that $I \succ J$. $\square$

We now complete the proof of the theorem. For every $A \in \mathcal{A}$, we restrict attention to

$$\mathcal{I}_A = \{I_A \mid I \in \mathcal{I}\}$$

and notice that $\succsim$ satisfies A1-A4 on this space, where A3, the unrestricted version of Archimedeanity, follows from the fact that all I_A vanish outside of the equivalence class A . Theorem 2 guarantees that existence of a nonnegative u_A that represents $\succsim$ on $\mathcal{I}_A$ via the product operation. The function u has to vanish outside of A . We define $u = \sum_{A \in \mathcal{A}} u_A$ which is well-defined because each p belongs to a unique A .²¹ It is straightforward to verify that $(\succsim \cdot, u)$ is a lexicarian representation of $\succsim$.

The proof of necessity of the axioms is straightforward and therefore omitted.

Clearly, we obtain a function u that can be separately re-scaled on each $A \in \mathcal{A}$, but is otherwise unique: there can be at most one $A \in \mathcal{A}$ consisting of p with $u(p) = 0$, where for $q \in A^c$ we have $u(q) > 0$. Further, for $q_1, q_2 \in A^c$, $u(q_1)/u(q_2)$ is uniquely defined. If we wish the relation $\succsim \cdot$ to have maximal equivalence classes, as in our construction, then it is unique. However, one may

²¹Alternatively, one could consider the restrictions of profiles to equivalence classes, obtain a function u_A that is defined only over A , and define u as their union.

have an equivalent representation with other relations $\succsim'$, which agree on the equivalence classes over which $u(q) > 0$ but may not consider all positions p with $u(p) = 0$ to be $\sim'$ -equivalent. $\square$

6.3 Proof of Proposition 1

It is straightforward to see that the axioms are necessary: given a representation of $\succsim$ by $V(I) = \sum_{\omega \in \Omega} pr(\omega) \sum_{t \in T} \delta^t \sum_{c \in C} I(c, t, \omega) \hat{u}(c)$, we have a representation as in Theorem 2 with $u(c, t, \omega) = pr(\omega) \delta^t \hat{u}(c)$. This means that, restricting attention to profiles in $\mathcal{I}_\omega$ or in $\mathcal{I}_{\omega'}$ preferences are identical if both ω, ω' are non-null (or if both are null, which means that $u(c, t, \omega) = u(c, t, \omega') = 0$ for all $(c, t) \in C \times T$). Similarly, A6 and A7 follow from the representation.

Conversely, assume that A5-A7 hold. Consider non-null $\omega, \omega' \in \Omega$ and their corresponding $\mathcal{I}_\omega, \mathcal{I}_{\omega'}$. Applying Theorem 2 to each of $\mathcal{I}_\omega, \mathcal{I}_{\omega'}$ separately, we conclude that there exist functions $u_\omega, u_{\omega'} : C \times T \rightarrow \mathbb{R}$ that represent $\succsim$ on these restricted spaces via the inner products (i.e., for $I, J \in \mathcal{I}_\omega$, $I \succsim J$ iff $I \cdot u_\omega \geq J \cdot u_\omega$, and similarly for $I, J \in \mathcal{I}_{\omega'}$). By A5, $\succsim$ on $\mathcal{I}_\omega$ is identical to its restriction to $\mathcal{I}_{\omega'}$. Thus, u_ω also represents $\succsim$ on $\mathcal{I}_{\omega'}$ (and vice versa). By the uniqueness result, $u_{\omega'}$ is a positive multiple of u_ω . Fix one non-null state ω_0 (if such exists) and for every ω' choose $\lambda_{\omega'} > 0$ such that $u_{\omega'} = \lambda_{\omega'} u_{\omega_0}$. That is, $u(c, t, \omega') = \lambda_{\omega'} u(c, t, \omega_0)$. If all states are null, u vanishes. Otherwise, normalize $(\lambda_{\omega'})_{\omega'}$ to be a probability vector pr .

Next consider two periods $t, t' \in T$ and, using similar reasoning applied to A6, conclude that, for every $t > 0$ there exists $\mu_t > 0$ such that $u(c, t, \omega_0) = \mu_t u(c, 0, \omega_0)$. Define $\hat{u}(c) = u(c, 0, \omega_0)$ and observe that $u(c, t, \omega) = pr(\omega) \mu_t \hat{u}(c)$. Finally, invoke A7 to conclude that, for $\delta \equiv \mu_1 > 0$ we have to have $\mu_t = \delta^t$ for all t . $\square$

6.4 Proof of Proposition 2

We start with necessity of the axioms. A8 follows from condition (ii). We first note that, for each $\sim^b$ -equivalence class A , A5-A7 hold for profiles whose

supports are contained in it, $\mathcal{I}_A$. This follows from the necessity of the axioms in Proposition 1. To see that these axioms hold in general, it remains to follow the lexicographic representation. For example, consider A5. Let there be given two non-null $\omega, \omega' \in \Omega$ and $I, J \in \mathcal{I}_\omega$, and $I', J' \in \mathcal{I}_{\omega'}$ with $I(c, t, \omega) = I'(c, t, \omega')$ and $J(c, t, \omega) = J'(c, t, \omega')$ for all $(c, t) \in C \times T$. We need to show that $I \succsim J$ iff $I' \succsim J'$. Assume that $I \succ J$. Let A be the $\succsim^b$ -maximal equivalence class such that $I_A \approx J_A$ (that is, $I_A \succ J_A$). Then, for any equivalence class $A' \succ^b A$ we have $I_{A'} \sim J_{A'}$ and hence $I'_{A'} \sim J'_{A'}$. Similarly, for A we have $I'_A \succ J'_A$ and thus $I' \succ J'$. If, however, $I \sim J$, we have to have $I_A \sim J_A$ for all A and thus also $I'_A \sim J'_A$ and $I \sim J$ follows. Similar reasoning applied to the necessity of A6 and A7.

To prove sufficiency of the axioms, we apply the sufficiency part of Proposition 1 for each $\sim^b$ -equivalence class A separately, and derive the representation as in (i) with $u(c, t, \omega) = pr_A(\omega) \delta_A^t \hat{u}(c)$ for all (c, t, ω) (where pr_A and δ_A are derived from Proposition 1). Finally, (ii) follows from A8. $\square$

7 Bibliography

- Adler, Matthew D. (2012), *Well-Being and Fair Distribution: Beyond Cost-Benefit Analysis*. Oxford University Press.
- Alcantud, Jose-Carlos R., Marco Mariotti, and Roberto Veneziani (2022), “Sufficientarianism”. *Theoretical Economics*, **17**: 1529-1557.
- Arrhenius, Gustaf (2000), “An Impossibility Theorem for Welfarist Axiologies”, *Economics and Philosophy*, **16**: 247-266.
- Asheim, Geir B. and Susumu Cato (2026), “The Dilemma between Repugnancy and Sadism Extends beyond Critical-Level Utilitarianism”, *Economics and Philosophy*, forthcoming.
- Blackorby, Charles, Walter Bossert, and David Donaldson (1995), “Intertemporal Population Ethics: Critical-Level Utilitarian Principles”, *Econometrica*, **65**: 1303-20.

- Blackorby, Charles, Walter Bossert, and David Donaldson (1997), “Critical-Level Utilitarianism and the Population Ethics Dilemma”, *Economics and Philosophy*, **13**: 197-230.
- Blackorby, Charles and David Donaldson (1984), “Social Criteria for Evaluation Population Change”, *Journal of Public Economics*, **25**: 13-33.
- Bloom, Paul (2016), *Against Empathy: The Case for Rational Compassion*. Ecco/Harper Collins.
- Blume, Lawrence, Adam Brandenburger, and Eddie Dekel (1991), “Lexicographic Probabilities and Choice under Uncertainty”, *Econometrica*, **59**: 61-79.
- Boonin, David (2014), *The Non-Identity Problem and the Ethics of Future People*, Oxford University Press.
- Bossert, Walter (1990), “Maximin Welfare Orderings with Variable Population Size”, *Social Choice and Welfare*, **7**: 39-45.
- Bossert, Walter, Susumu Cato, and Kohei Kamaga (2023), “Thresholds, Critical Levels, and Generalized Sufficientarian Principles”, *Economic Theory*, **75**: 1099-1139.
- Broome, John (2004), *Weighing Lives*, Oxford University Press.
- Brickman, Philip, Donald T. Campbell (1971), “Hedonic Relativism and Planning the Good Society”, in M. H. Appley (Ed.), *Adaptation Level Theory: A Symposium*. Academic Press.
- Brickman, Philip, Dan Coates, and Ronnie Janoff-Bulman (1978), “Lottery Winners and Accident Victims: Is Happiness Relative?” *Journal of Personality and Social Psychology*, **36**: 917-927.
- Carlson, Erik (1998), “Mere Addition and the Two Trilemmas of Population Ethics”, *Economics & Philosophy*, **14**: 283-306.
- de Finetti, Bruno (1931), “Sul Significato Soggettivo della Probabilità”, *Fundamenta Mathematicae*, **17**: 298-329.

- de Finetti, Bruno (1937), “La Prévision: Ses Lois Logiques, Ses Sources Subjectives”, *Annales de l’Institut Henri Poincaré*, **7**: 1-68.
- Dhillon, Amrita, and Jean-François Mertens (1999), “Relative utilitarianism”, *Econometrica*, **67**(3), 471-498.
- Diener, Ed (1984), “Subjective Well-being”, *Psychological Bulletin*, **95**: 542-575.
- Easterlin, Richard A. (1973), “Does money buy happiness?” *The Public Interest*, **30**: 3-10.
- Easterlin, Richard A. (1974), “Does Economic Growth Improve the Human Lot? Some Empirical Evidence”, in *Nations and Households in Economic Growth*. P. A. David and M. W. Reder, Academic Press: 89-125.
- Easterlin, Richard A. (2017), “Paradox Lost?”, *Review of Behavioral Economics*, **4**: 311-339.
- Frankfurt, Harry G. (1987), “Equality as an Ideal”, *Ethics*, **98**: 21-43.
- Frick, Johann (2020), “Conditional Reasons and the Procreation Asymmetry”, *Philosophical Perspectives*, **34**: 53-87.
- Gilboa, Itzhak (2009), *Theory of Decision under Uncertainty*. Econometric Society Monograph Series Vol. No. 45. Cambridge: Cambridge University Press.
- Gilboa, Itzhak and Larry Samuelson (2022), “What Were You Thinking? Revealed Preference Theory as Coherence Test”, *Theoretical Economics*, **17**, 507-519.
- Gustafsson, Johan E. (2022), “Our intuitive grasp of the repugnant conclusion.” *The Oxford Handbook of Population Ethics*, 371-89.
- Hammond, Peter J. (1976), “Equity, Arrow’s Conditions, and Rawls’ Difference Principle”, *Econometrica*, **44**: 793-804.
- Harsanyi, John C. (1955), “Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparisons of Utility,” *Journal of Political Economy*, **63**: 309-321.

- Helson, Harry (1947), "Adaptation-Level as Frame of Reference for Prediction of Psychophysical Data", *American Journal of Psychology*, **60**: 1-29.
- Helson, Harry (1948), "Adaptation-Level as a Basis for a Quantitative Theory of Frames of Reference", *Psychological Review*, **55**: 297-313.
- Huemer, Michael (2008), "In Defence of Repugnance", *Mind*, **117**(468): 899-933.
- Hume, David (1739-40), *A Treatise of Human Nature*. Edited by L. A. Selby-Bigge and P. H. Niddich. Oxford: Clarendon Press, 1978.
- Hume, David (1751), *An Enquiry Concerning the Principles of Morals*. Edited by Tom L. Beauchamp. Oxford: Oxford University Press, 1998.
- Hurka, Thomas (1983), "Value and Population Size", *Ethics*, **93**: 496-507.
- Kahneman, Daniel, Alan B. Krueger, David A. Schkade, Norbert Schwarz, and Arthur A. Stone (2004), "A Survey Method for Characterizing Daily Life Experience: The Day Reconstruction Method", *Science*, **306**: 1776-1780.
- Kauppinen, Antti (2013), "Meaning and Happiness", *Philosophical Topics*, **41**: 161-185.
- Krantz, David H., R. Duncan Luce, Patrick Suppes, and Amos Tversky (1971), *Foundations of Measurement: vol. I*. Academic Press, New York.
- Li, P. F. Jonah, Y. Joel Wong, and Ruth C.-L. Chao (2019), "Happiness and Meaning in Life: Unique, Differential, and Indirect Associations with Mental Health", *Counselling Psychology Quarterly*, **32**: 1-19.
- Lucas, Richard E., Portia S. Dyrenforth, and Ed Diener (2008), "Four Myths about Subjective Well-Being", *Social and Personality Compass*, **2**: 2001-2015.
- Mancini, Anthony D., George A. Bonanno, and Andrew E. Clark (2011), "Stepping Off the Hedonic Treadmill: Individual Differences in Response to Major Life Events", *Journal of Individual Differences*, **32**: 144-152.
- Maslow, Abraham H. (1943), "A Theory of Human Motivation", *Psychological Review*, **50**: 370-396.

- Mill, John Stuart. (1863), *Utilitarianism*. Chapter 2.
- Narveson, Jan (1973), “Moral Problems of Population”, *The Monist*, **57**: 62-86.
- Nebel, Jacob (2024), “Extensive Measurement in Social Choice”, *Theoretical Economics*, **19**: 1581-1618.
- Osborne, Martin J. (2025), *Models in Political Economy*. OpenBook Publishers, Cambridge, UK.
- Parfit, Derek (1976), “On Doing the Best for Our Children”, in: M. Bayles. ed.. *Ethics and Population* (Schenkman, Cambridge).
- Parfit, Derek (1982), “Future generations: Further problems”, *Philosophy and Public Affairs*, **11**: 113-172.
- Parfit, Derek (1984), *Reasons and Persons*. Oxford University Press.
- Parfit, Derek (1997), “Equality and Priority”, *Ratio*, **10**: 202-221.
- Persson, Ingmar (2004), “The Root of the Repugnant Conclusion and its Rebuttal”, in J. Ryberg and T. Tannsjö (eds.) *The Repugnant Conclusion*, Library of Ethics and Applied Philosophy: 187-200.
- Pivato, Marcus (2020), “Rank-additive population ethics”, *Economic Theory*, **69**: 861-918.
- Rachels, Stuart (2004), “Repugnance or Intransitivity: A Repugnant but Forced Choice”, in J. Ryberg and T. Tannsjö (eds.) *The Repugnant Conclusion*, Library of Ethics and Applied Philosophy: 163-186.
- Rawls, John A. (1971), *A Theory of Justice*. Harvard University Press.
- Roberts, Melinda A. (2011), “The Asymmetry: A Solution ”, *Theoria*, **77**: 333-367.
- Savage, Leonard J. (1954), *The Foundations of Statistics*. New York: John Wiley and Sons. (Second addition in 1972, Dover)
- Schwarz, Norbert and Gerald L. Clore (1983), “Mood, Misattribution, and Judgments of Well-Being: Informative and Directive Functions of Affective States”, *Journal of Personality and Social Psychology*, **45**: 513-523.

- Sikora, Richard I. (1975), "Utilitarianism: The Classical Principle and the Average Principle". *Canadian Journal of Philosophy*, **5**: 409-419.
- Smith, Adam (1759), *The Theory of Moral Sentiments*. Edited by D. D. Raphael and A. L. Macfie. Indianapolis: Liberty Fund, 1982.
- Strack, Fritz, Leonard L. Martin, and Norbert Schwarz (1988), "Priming and Communication: Social Determinants of Information Use in Judgments of Life Satisfaction", *European Journal of Social Psychology*, **18**: 429-442.
- Tännsjö, Torbjörn (2002), "Why We Ought to Accept the Repugnant Conclusion", *Utilitas*, **14**: 339-350.
- Temkin, Larry S. (1987), "Intransitivity and the Mere Addition Paradox", *Philosophy and Public Affairs*, **16**: 138-187.
- Temkin, Larry S. (2012), *Rethinking the Good: Moral Ideals and the Nature of Practical Reasoning*. Oxford University Press.
- Veenhoven, Ruut and Floris Vergunst (2014), "The Easterlin Illusion: Economic Growth Does Go with Greater Happiness", *International Journal of Happiness and Development*, **1**: 311-343.
- von Neumann, John and Oskar Morgenstern (1944), *Theory of Games and Economic Behavior*. Princeton, N.J.: Princeton University Press.
- Zuber, Stéphane and Geir B. Asheim (2012), "Justifying Social Discounting: The Rank-Discounted Utilitarian Approach", *Journal of Economic Theory*, **147**: 1572-1601.